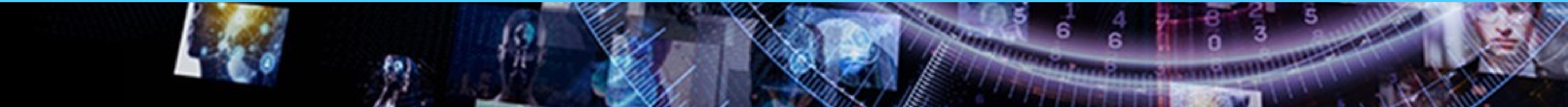
C.A.S.A. Course

CONTEXT AWARE SECURITY ANALYTICS IN COMPUTER VISION

proff. Michele Nappi, Fabio Narducci and Carmen Bisogni

aa.2024/2025



Lecturers

Prof. **Michele Nappi**
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Prof. **Fabio Narducci**
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Dr. **Carmen Bisogni**
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The organization of the CASA Course

- **The official schedule**:
  - Thursday: 11am – 1pm (Lab Hopper): mainly dedicated to theoretical lectures
  - Friday: 1,30pm – 4.30pm (Lab Hopper): mainly dedicated to laboratory activities and practise

  But, due to some national holidays we need to integrate with Monday lectures. All details about that are notified through the e-learning platform.

- Students are invited to attend regularly the lessons.
- First weeks are intended to transfer the theoretical topics and to be trained on the use of library and tools for Machine/Deep Learning models and experiments.
- In a second part of the course, lab activities and exercises will be assigned to get practise with the tools and the problems proposed by the course.

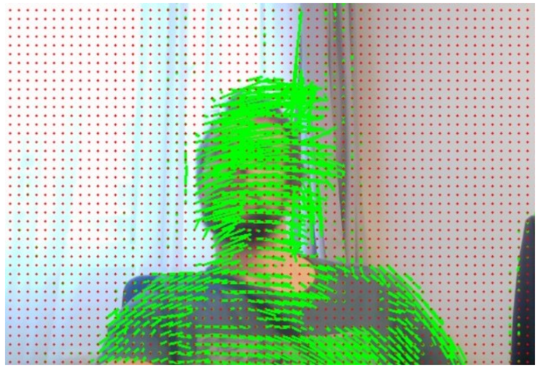
Examination

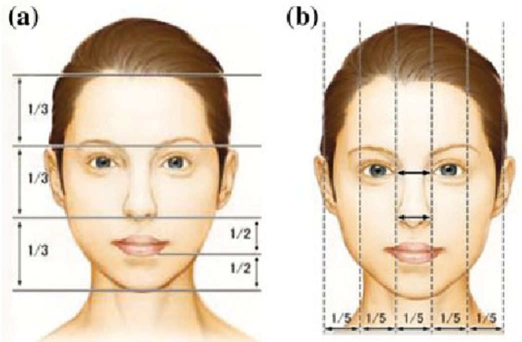
- To complete the course the student must demonstrate to know the theory of automatic learning and be aware of the potentials and the limitations of the solution at the state of the art.
- The examination is organized in two parts:

1. **Approximately in the middle of the course**
    - a written exam takes place. It consists in a quiz with multiple choice and open-ended questions.
    - it is meant to assess the expertise of the candidate with machine learning theory and applications
    - Candidates who pass the written quiz may choose to work on a deep learning project.
2. **At the end of the course**
    - Those who have passed the written quiz and those who have elected to work on the project will be required to present and discuss the assigned deep learning project during one of the scheduled examination sessions.
    - Those who have not passed the written quiz, or who, having passed the quiz, have declined to work on the project, will be required to take a written quiz and participate in an oral discussion on all topics of the CASA course.

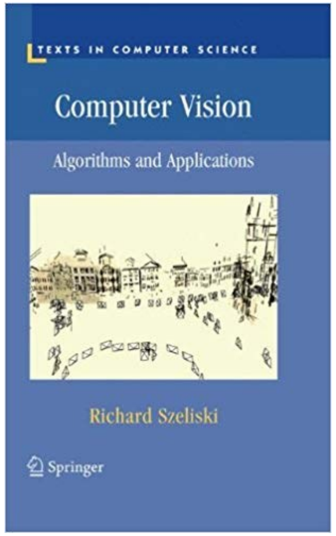
Let’s have a look at the (tentative) syllabus

- Introduction to Computer Vision
- Image Features and Matching
- Image Segmentation
- Image Motion
- Tracking
- Scene Recognition and Understanding
- Face Analysis
- Neural Network and Deep Learning in Computer Vision

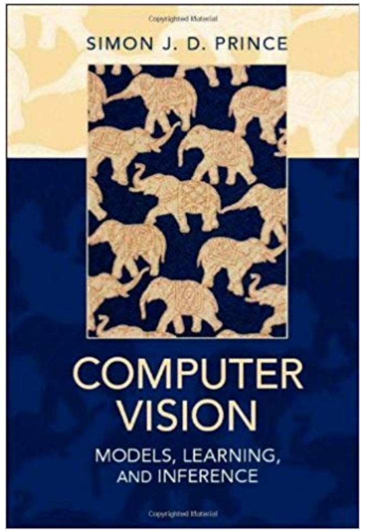


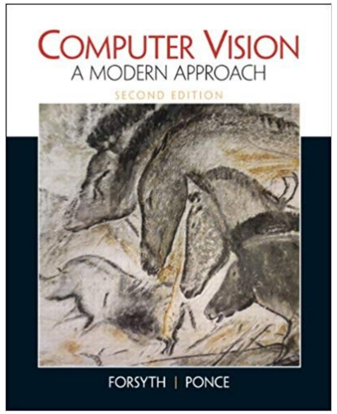
Books



Szeliski, R. (2010). Computer vision: algorithms and applications. Springer Science & Business Media.

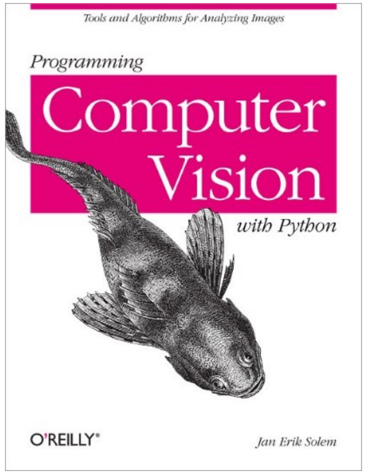
Prince, S. J. (2012). Computer vision: models, learning, and inference. Cambridge University Press.





Forsyth, D. A., & Ponce, J. (2002). Computer vision: a modern approach. Prentice Hall Professional Technical Reference.

Solem, Jan Erik (2012). Programming Computer Vision with Python: Tools and algorithms for analyzing images. " O'Reilly Media, Inc.".



What is computer vision?

Computer Vision definition

"The field of computer vision is providing an inspiration for researchers, scientists and engineers to develop systems that ***mimic*** *the incredible and complex* ***human vision capabilities*** *and, in some case, <u>surpass them</u>."*

"It is important for everyone to understand and appreciate the roles that computer vision plays in our society."

"Concepts of early vision, image processing, segmentation, camera geometry and motion, character recognition, deep learning and a much more."

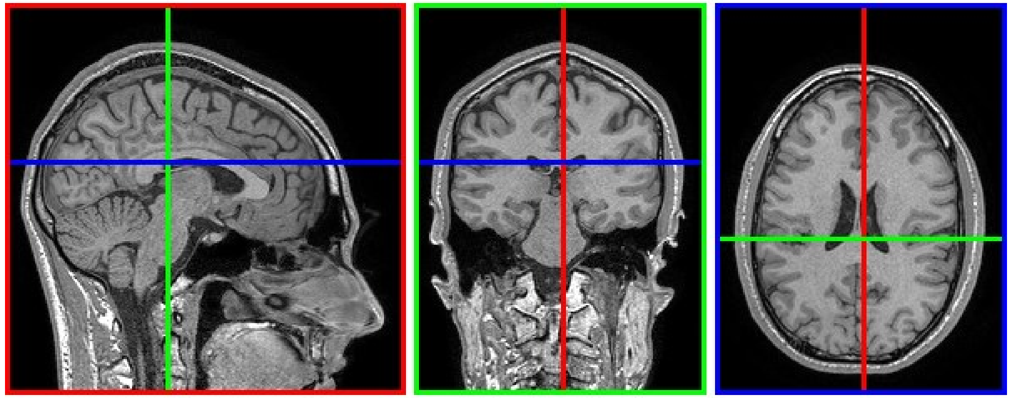
Junsong Yuan, Associate Professor in Computer Science and Engineering at the University at Buffalo.

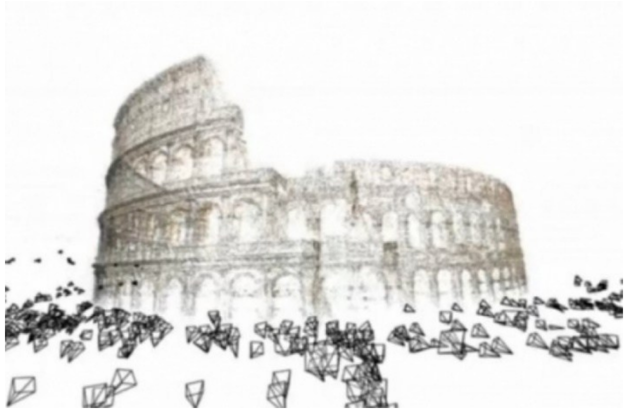
What is computer vision?

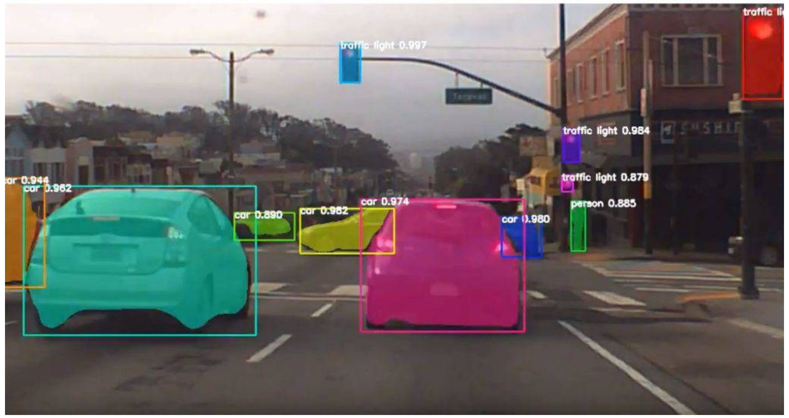
- What is the nature of computation involved in visual tasks?
- How might we build machines that **can see**?

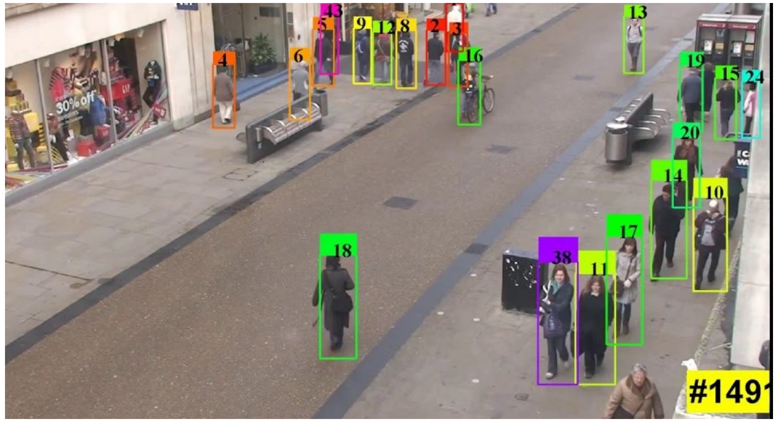
- Computer vision has a dual goal:
  - From the ***biological science point of view***, computer vision aims to come up with computational models for human visual system.
  - From the ***engineering point of view***, computer vision aims to build autonomous systems to perform some of the tasks which the human visual system can perform and even surpass it in many cases.
- *The two goals are strongly and intimately related.*

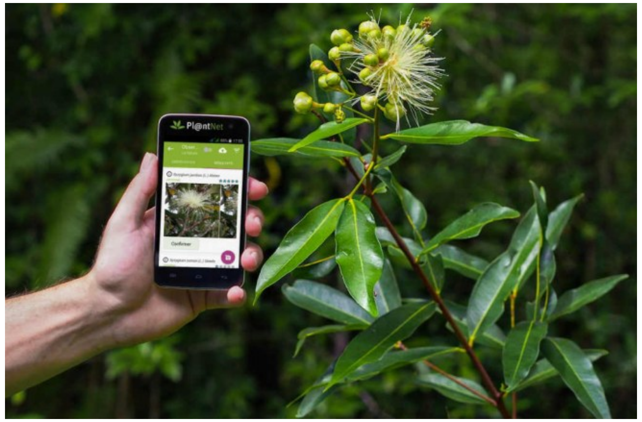
Why computer vision is so interesting?



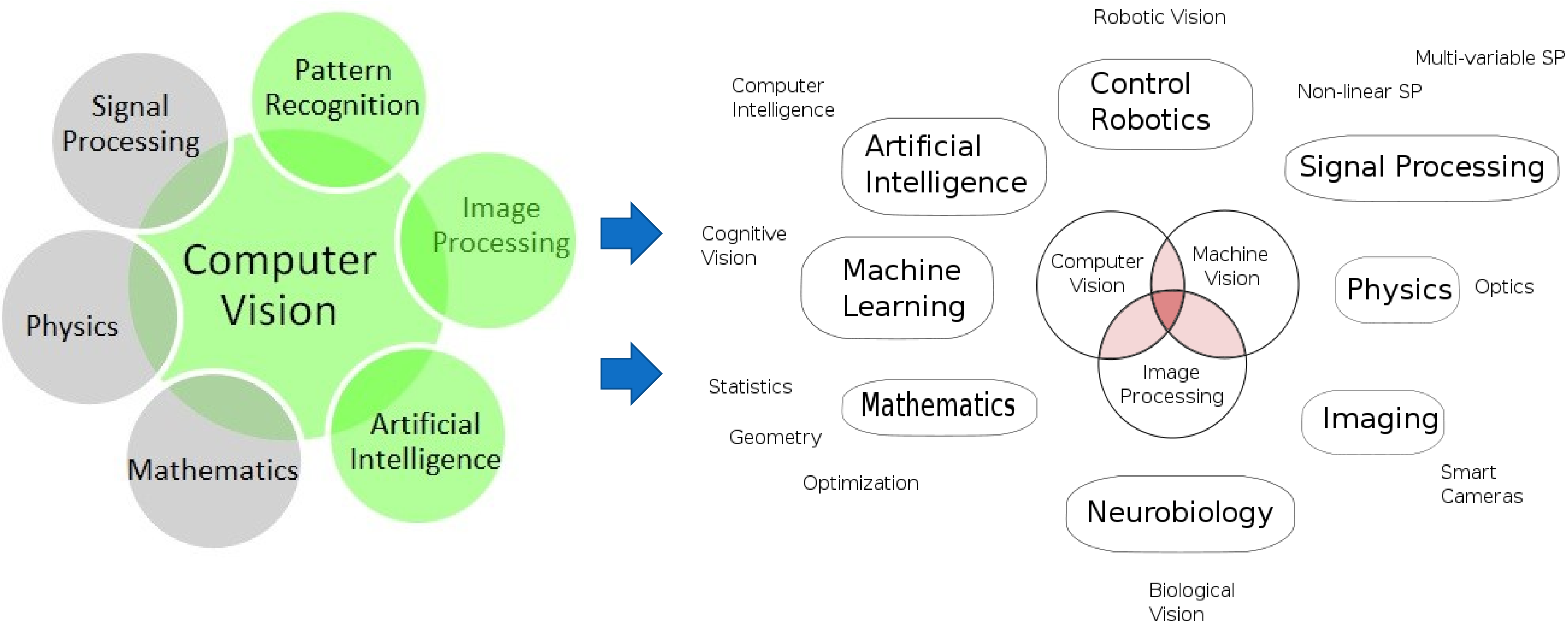




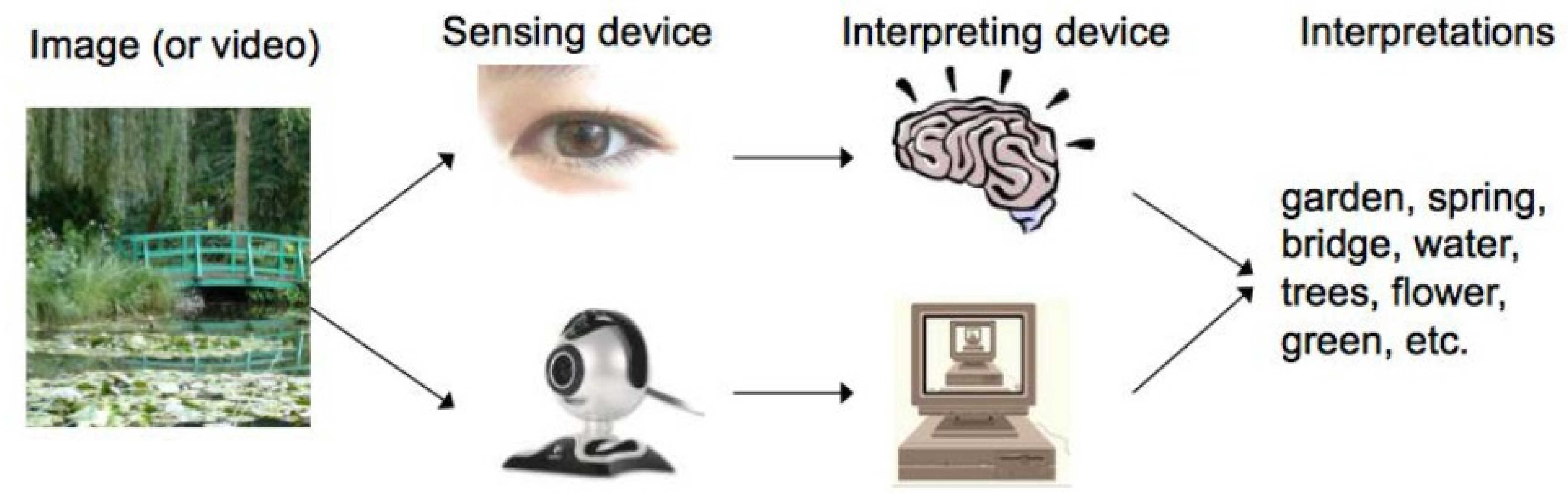




...but why it is also so difficult?



The goal of Computer Vision



Talking about human capabilities

- Vision contributes to «predict»
- Vision contributes to «decide»
- Vision contributes to «take action»
- …
- Vision contributes a lot!!!!

Ok, but what do computers see?

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

Ok, but what do computers see?

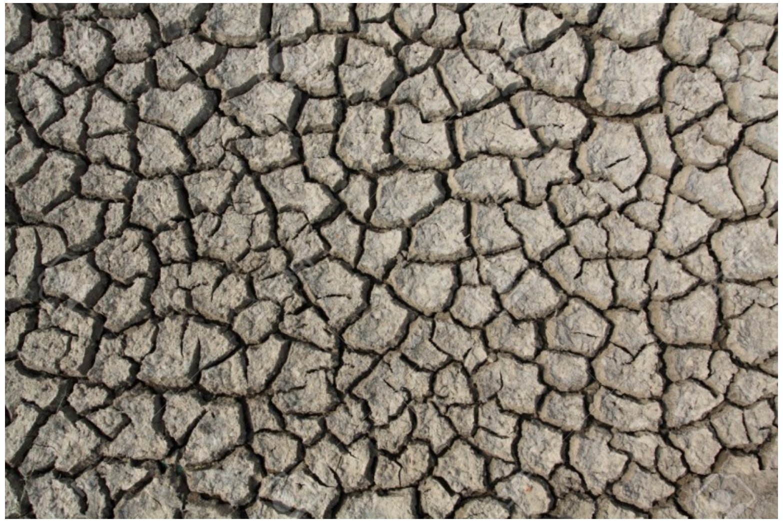
NUMBERS		
R 255 G 0 B 0	R 102 G 102 B 255	R 51 G 204 B 153
R 255 G 255 B 102	R 255 G 0 B 204	R 51 G 204 B 255
R 51 G 51 B 0	R 51 G 51 B 153	R 255 G 153 B 153

Are human eyes foolproof?

What is this?

Context!!!

Elephant



Dry soil

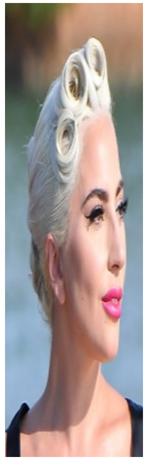
How does cognitive recognition work?

- What is this picture about?
- What is the subject?
- Are thay happy or angry?
- Did they win something?
- How can you relate all those things together?

How does cognitive recognition work?

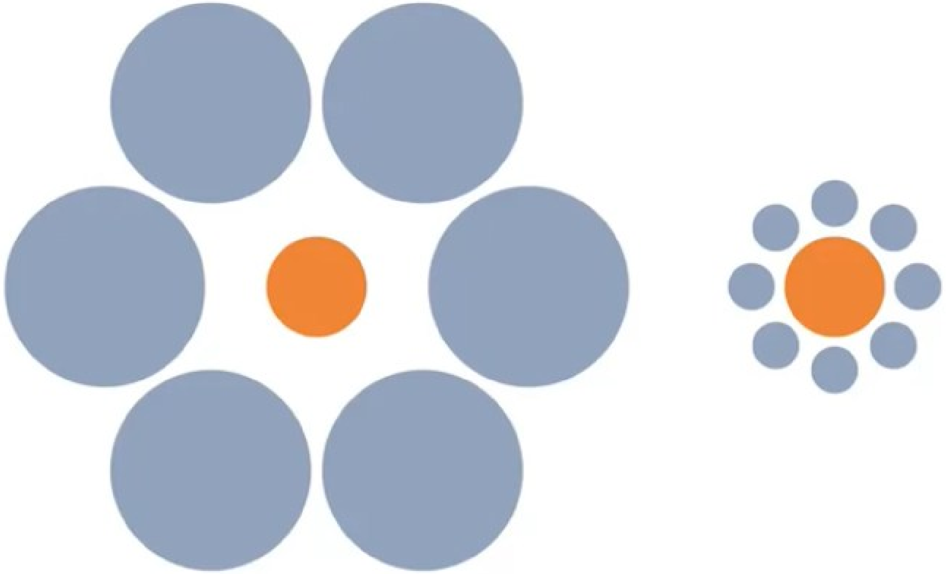
Who is in the picture?

And this?



If we knew the secrets of human vision…

- If we would be able to build a human-inspired electronic eye did we resolve all our issues?
- Answer is….NO!
- Human eyes are wonderful complex but not foolproof at all.
- What about the red cirles?



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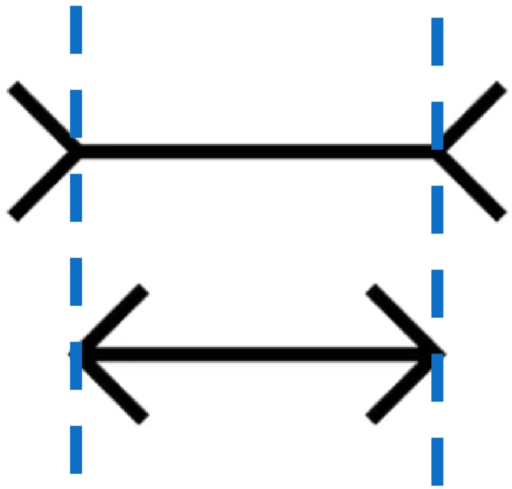
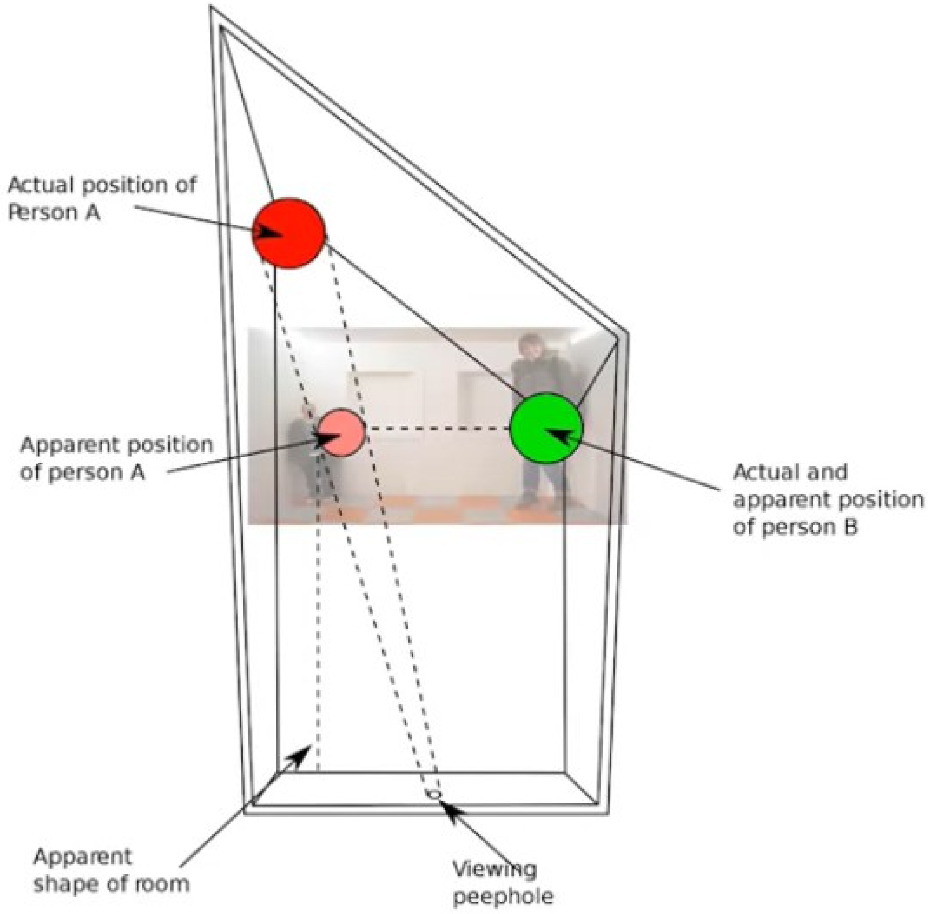
Visual illusions

The Muller-Lyer illusion

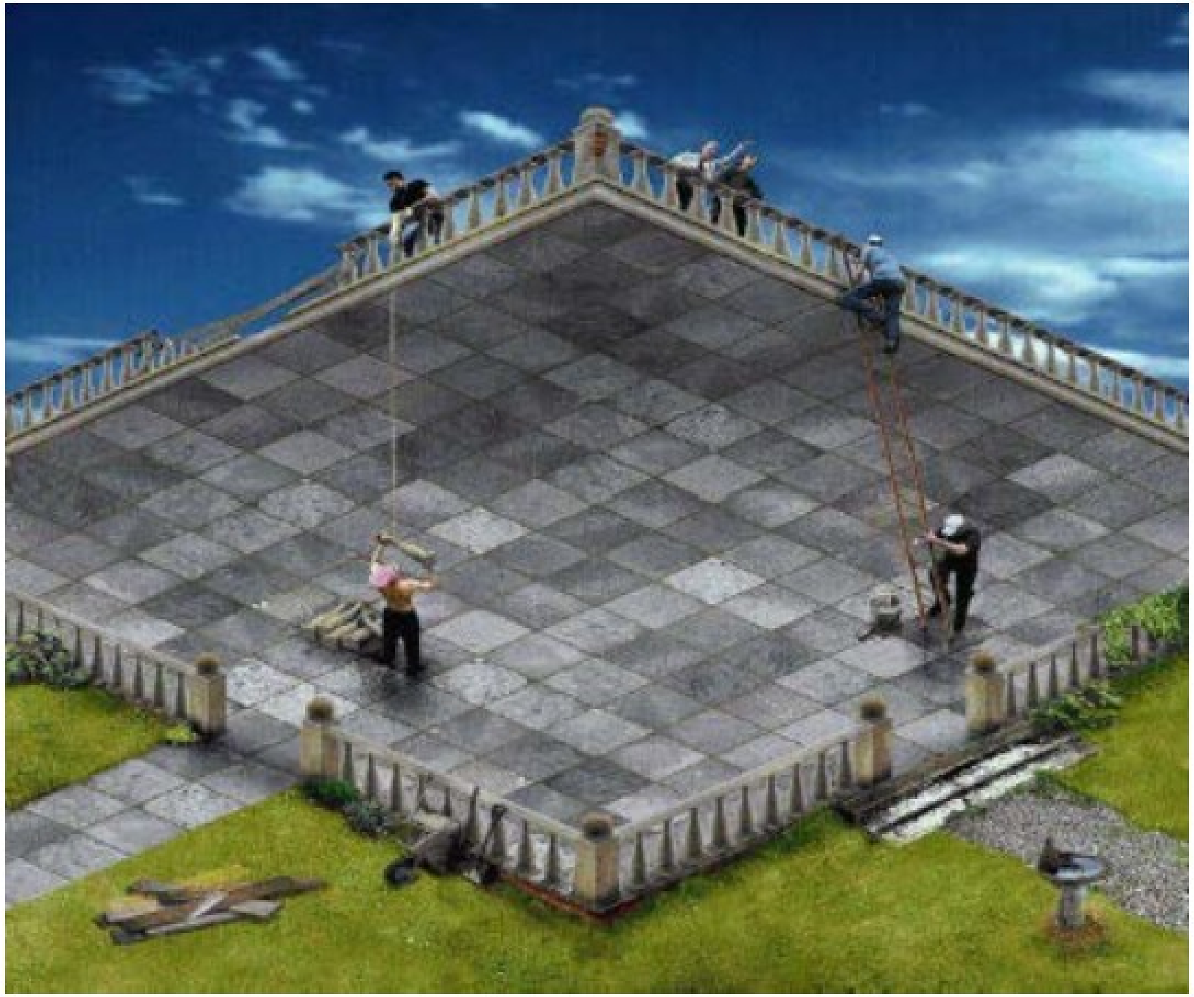


Visual illusions

Ames room

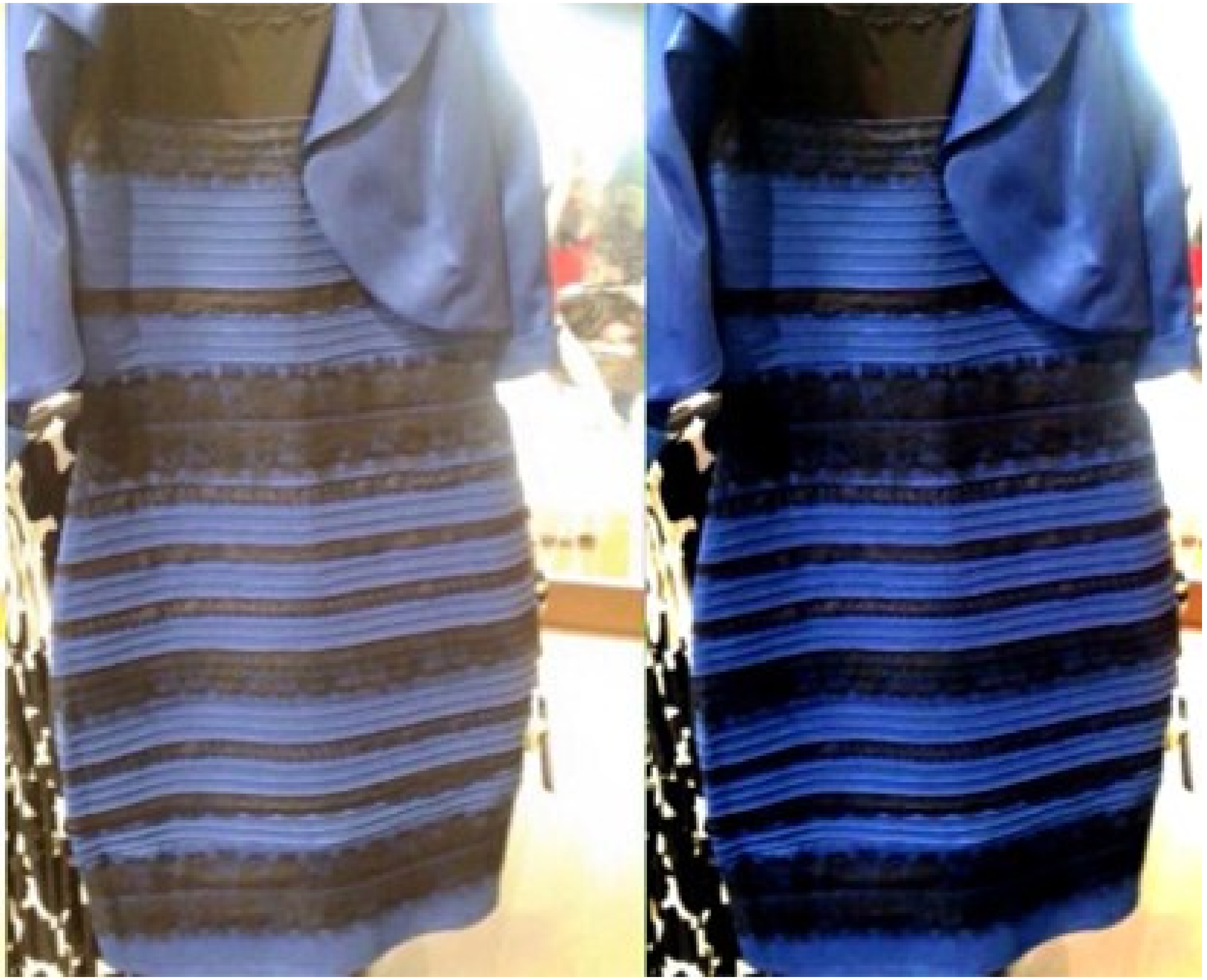


Visual illusions



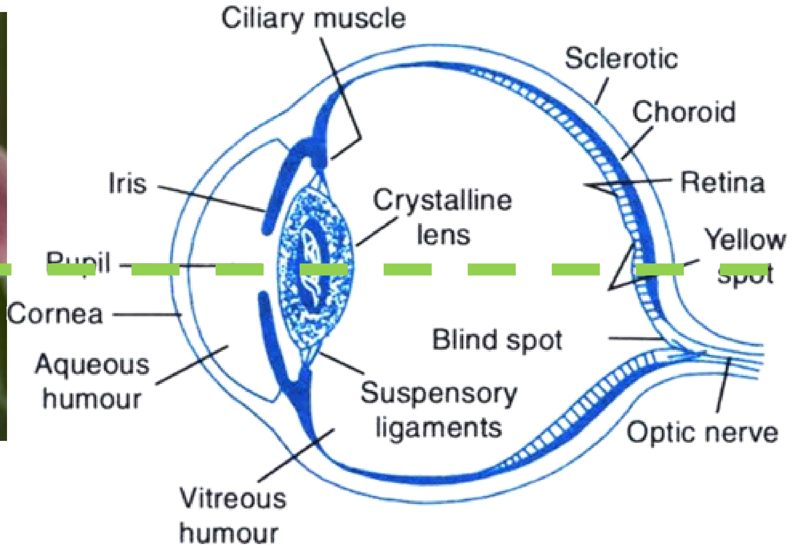
Very recent visual illusion

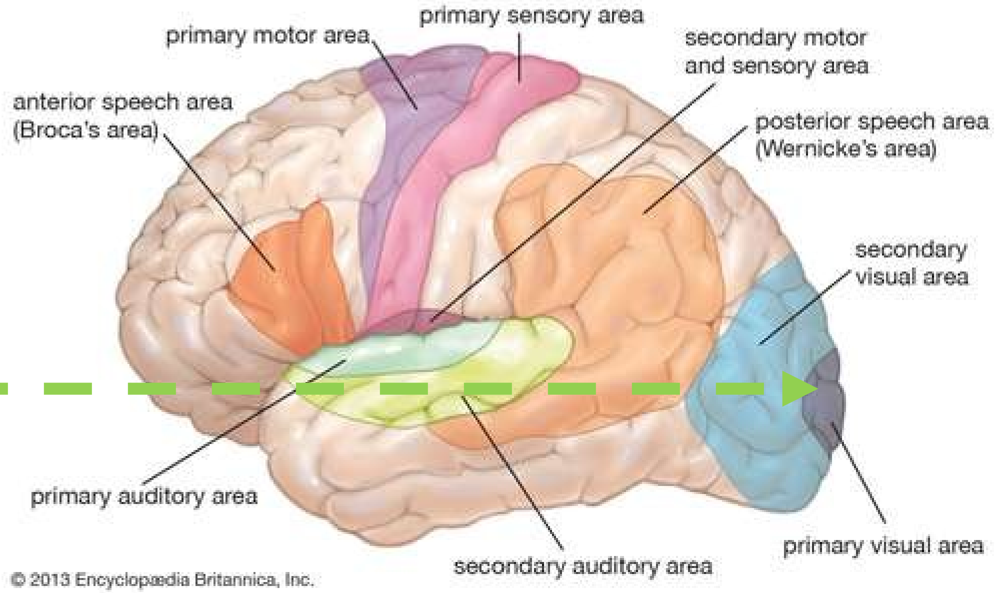
- Which color is this dress?



When observing an object…

The object | **The eye** | **The brain**





Is in the scene

We know how it works

What does it happen here?

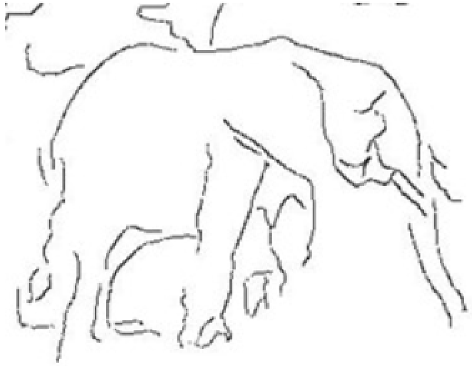
Low-Mid-High-level vision

- The vision can be ideally classified into three different levels:
  - **Low level vision (image processing, pixels, edges, corners and so on):**
    - Is concentrated in discovering what information about the world can be initially extracted from the image.
  - **Mid level vision (segmentation, pattern and motion analysis):**
    - In contrast to low-level vision, which recovers information from the sensors, mid-level vision concerns how this information gets organized into what we experience as objects and surfaces. ("Mid-level vision" is nearly coextensive with what is called "perceptual organization.")
  - **High level vision (Object recognition and understanding):**
    - processes complete the job of delivering a coherent interpretation of the image. It is assumed that the low-level processes make an adequate segmented representation of the two- and three-dimensional structure of the image. The high-level processes must determine **what objects are present in the scene** and interpret their **interrelations**. Assisting the low-level processes through a top-down flow of information, the high-level processes also help deciding what is present in an image.

<u>Low</u>-Mid-High-level vision

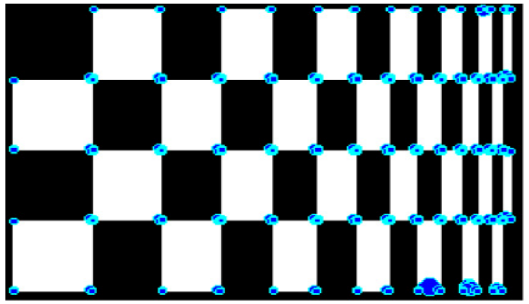
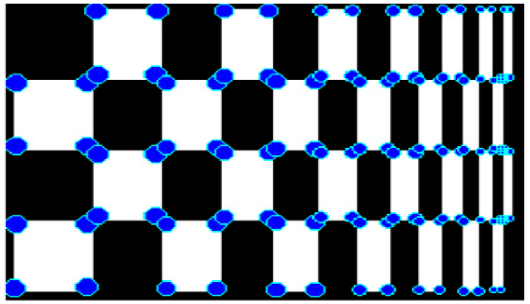
Low-level Contour Detection

- Use **Pb** (probability of boundary) as input
  - Combines local brightness, texture and color cues
  - Trained from human-marked segmentation boundaries
  - Outperforms existing local boundary detectors including Canny
- Canny's hysteresis thresholding

[Martin, Fowlkes & Malik 02]



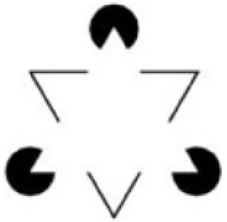
Low-**Mid**-High-level vision

Mid-level Vision

- What is Mid-level Vision?
  - It’s not low-level vision (e.g. edge detection)
  - It’s not high-level vision (e.g. object recognition)

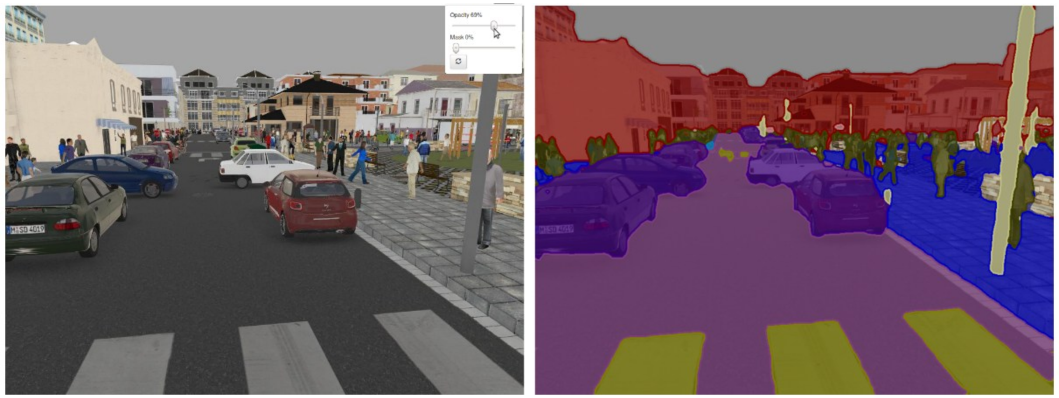
- Key Problems in Mid-level Vision

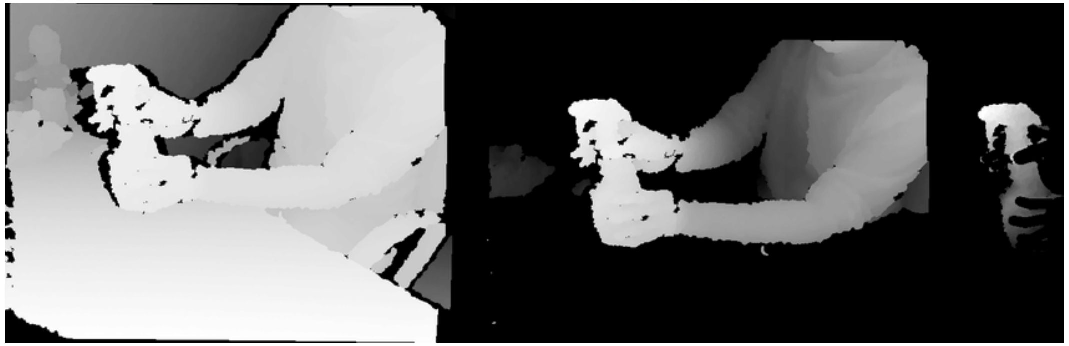
region grouping
/image segmentation



contour grouping
/visual completion

figure/ground
organization

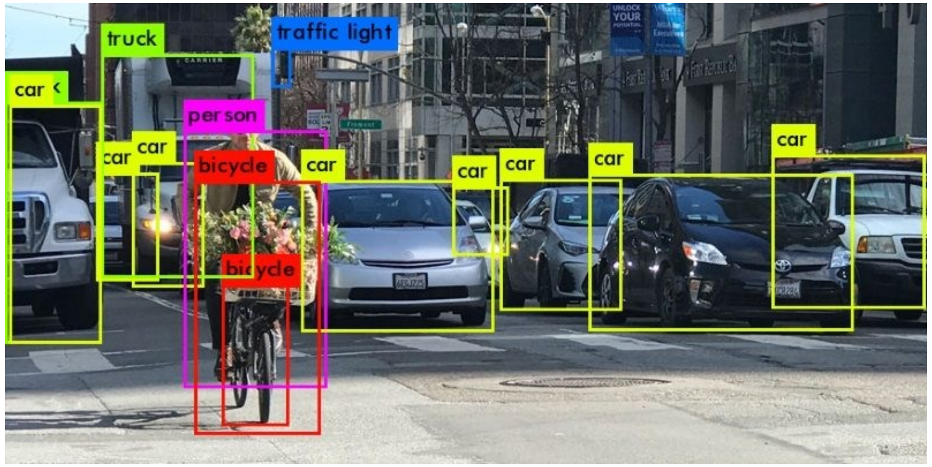


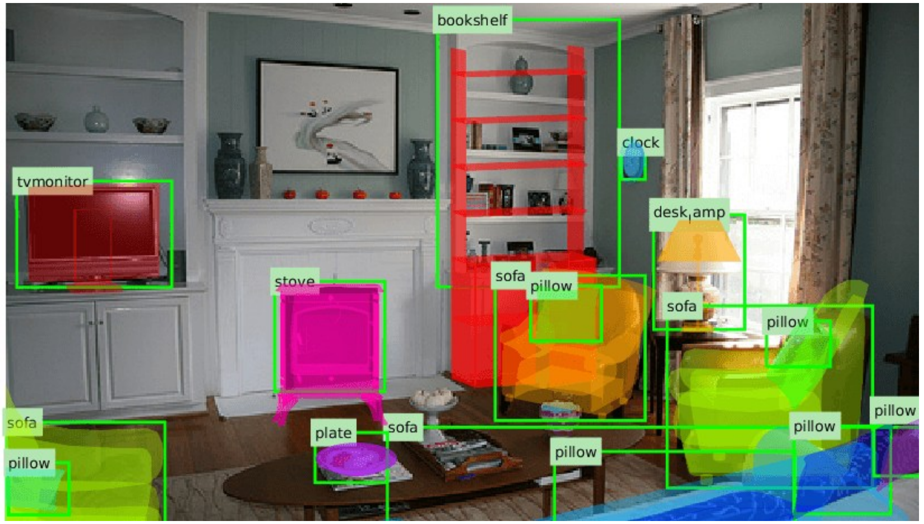


Low-Mid-**High**-level vision

High-Level Computer Vision

- Detection of classes of objects (faces, motorbikes, trees, cheetahs) in images
- Recognition of specific objects such as George Bush or machine part #45732
- Classification of images or parts of images for medical or scientific applications
- Recognition of events in surveillance videos
- Measurement of distances for robotics





What can we conclude so far?

Computer vision is a hard task!

Brief history of Computer Vision

Little story about computer vision

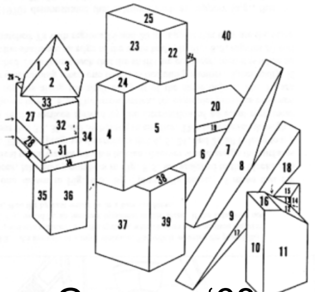
"In 1966, Marvin Minsky at MIT asked his undergraduate student Gerald Jay Sussman to "*spend the summer linking a camera to a computer and getting the computer to <u>describe</u> what it saw*".

We now know that the problem is slightly more difficult than that."

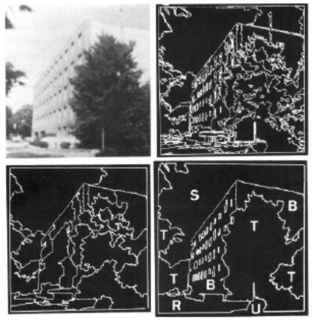
(*Szeliski 2009, Computer Vision*)

Ridiculously brief history of computer vision

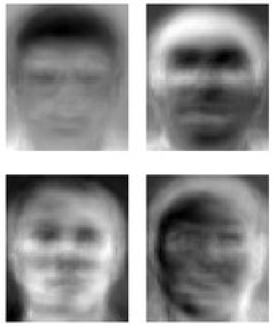
- 1966: Minsky assigns computer vision as an undergrad summer project
- 1960’s: interpretation of synthetic worlds
- 1970’s: some progress on interpreting selected images
- 1980’s: ANNs come and go; shift toward geometry and increased mathematical rigor
- 1990’s: face recognition; statistical analysis in vogue
- 2000’s: broader recognition; large annotated datasets available; video processing starts
- Today: the challenge is scene understanding and context awareness.



Guzman ‘68



Ohta Kanade ‘78

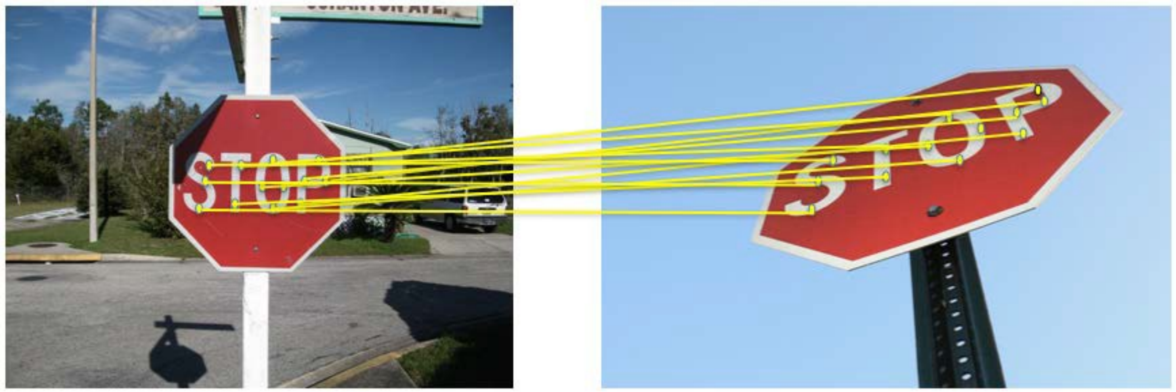


Turk and Pentland ‘91

The last 20 years: Normalised cut

Shi & Malik, 1997

The last 20 years: SIFT & Object recognition



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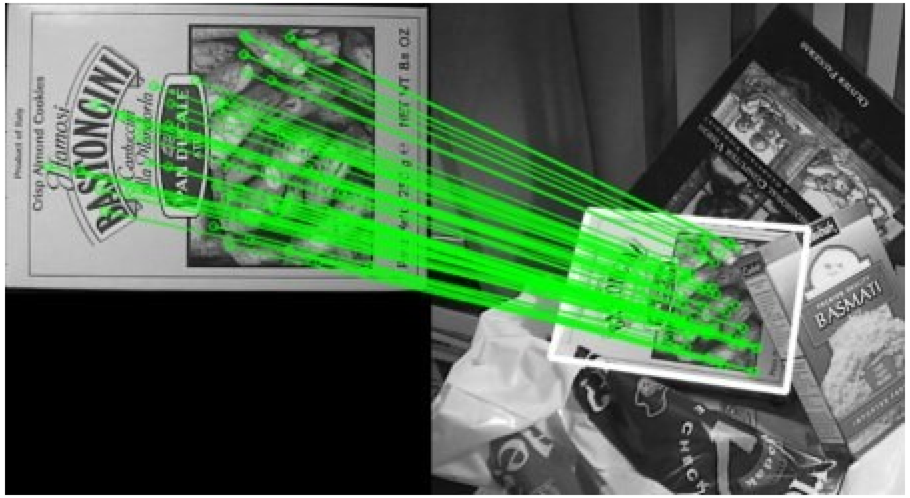
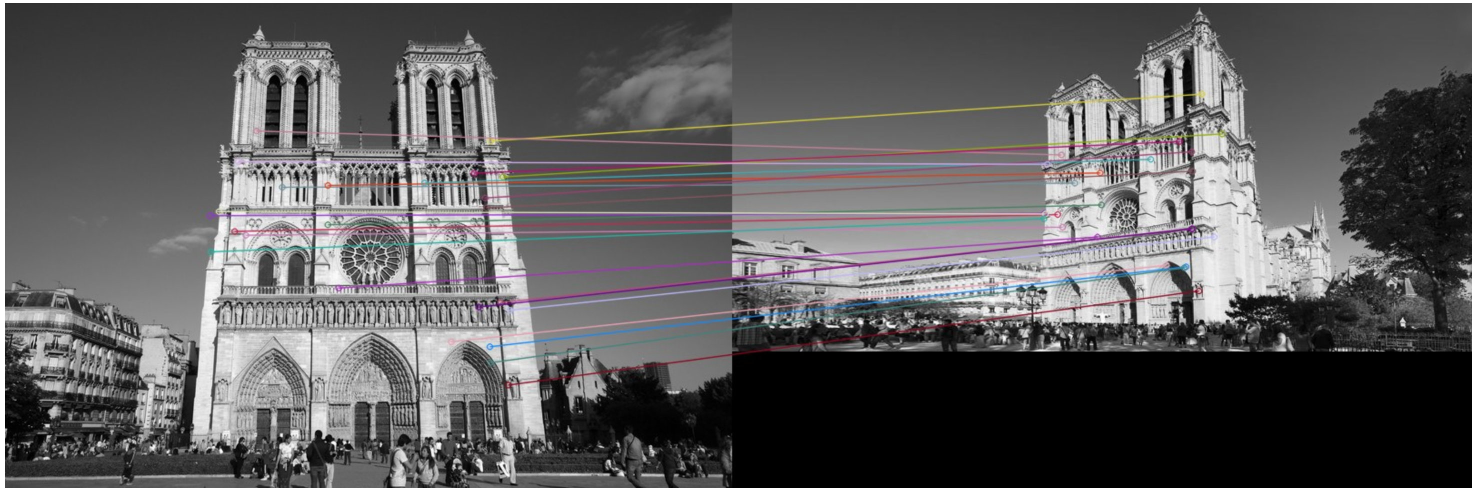
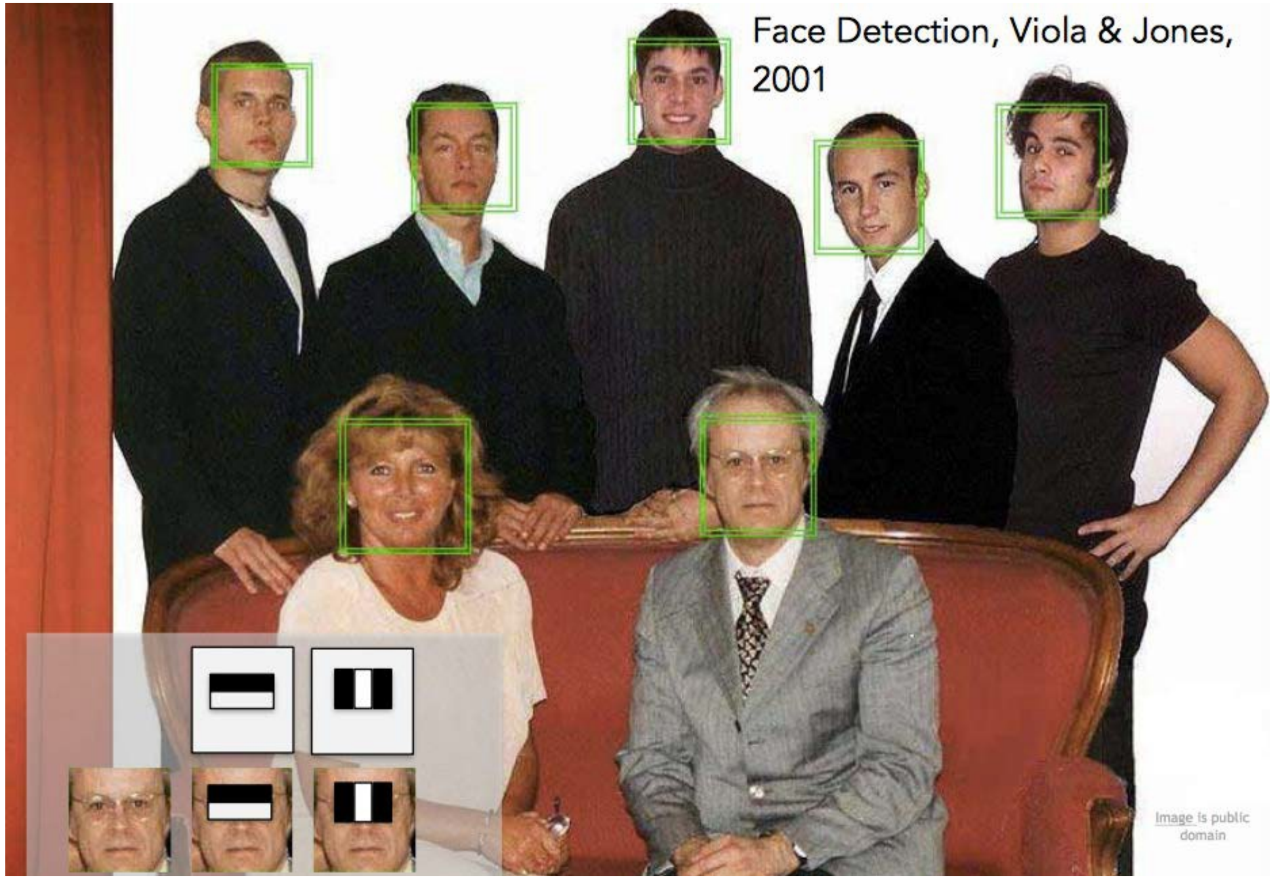


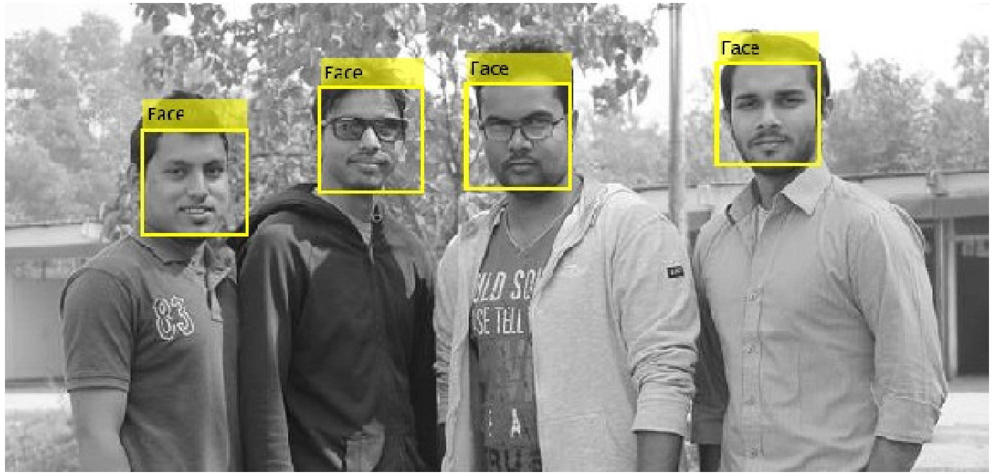


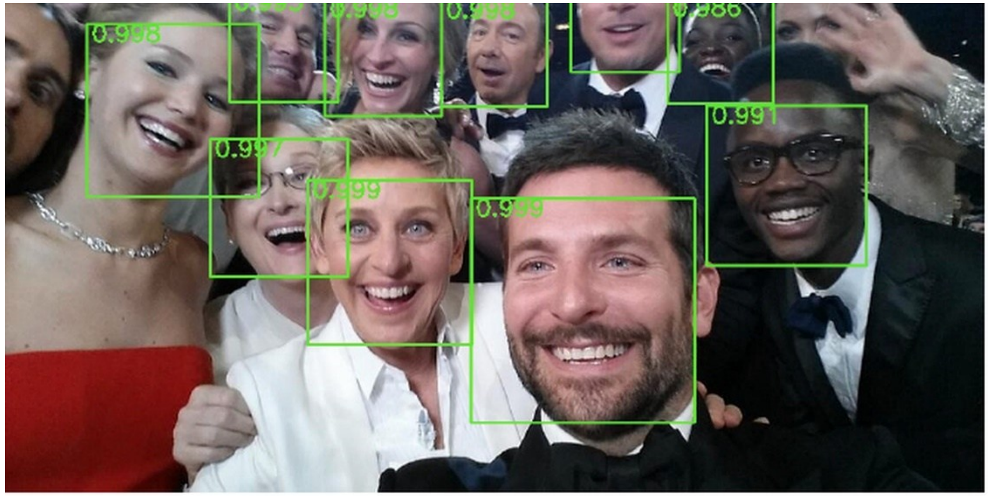
David Lowe, 1999

The last 20 years: Face Detector

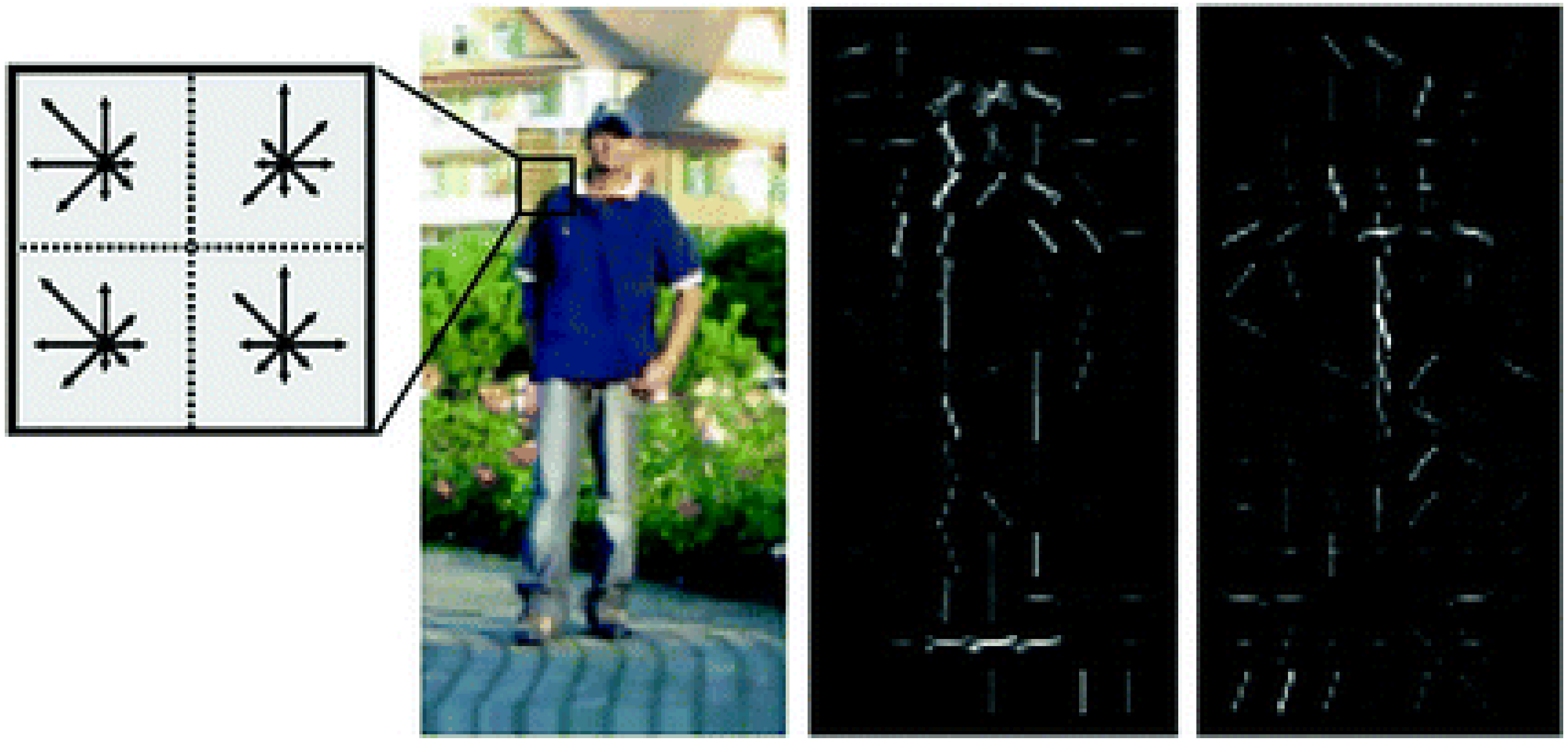


Viola & Jones, 2001





The last 20 years: HoG

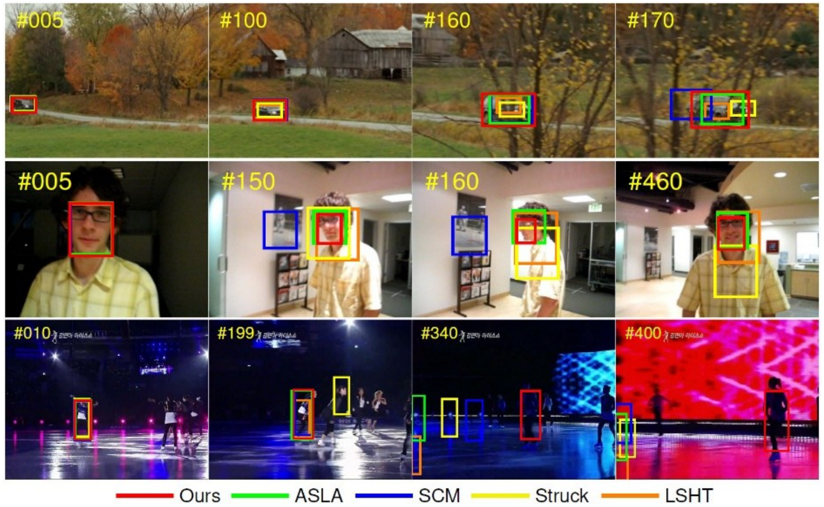


Histogram of Oriented Gradients, Dalal & Triggs, 2005

Felzenswalb et al., Deformable part model, 2009

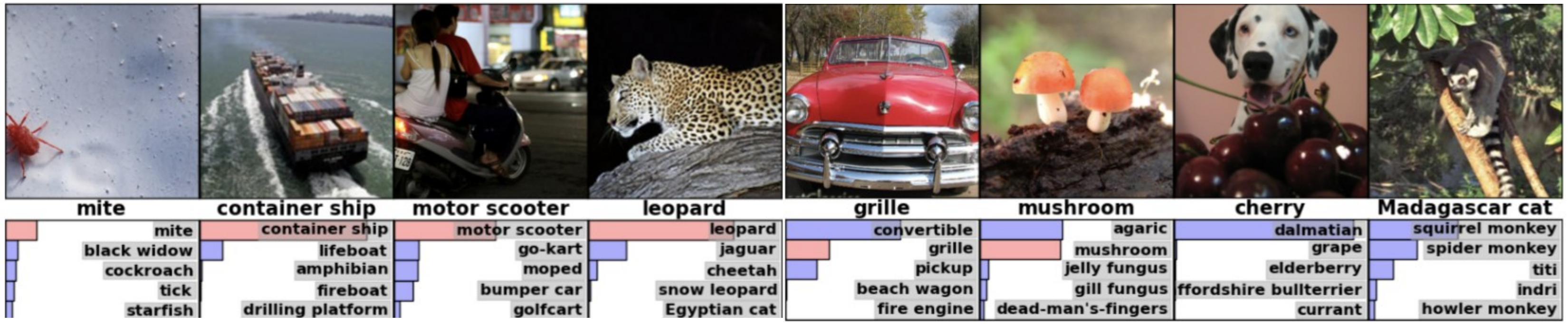
The last 20 years: Competitions!

Visual Object Tracking (VOT)



DensePose

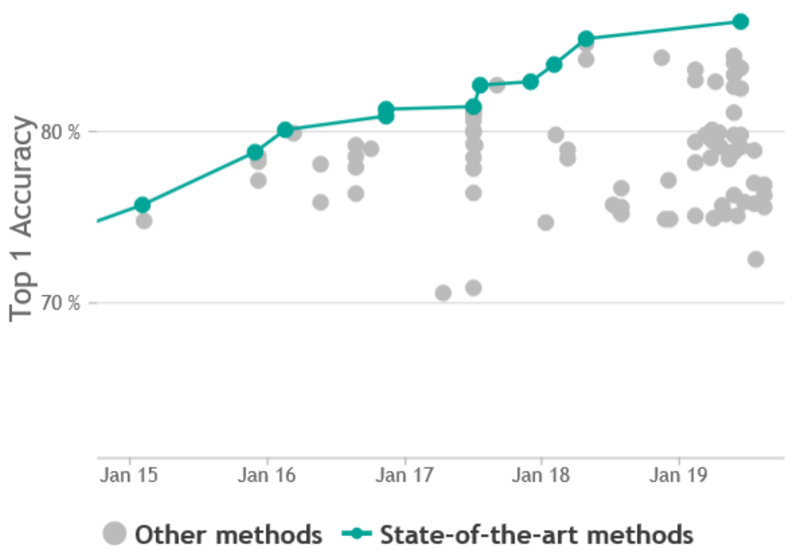
ImageNet Challenge



ImageNet (Fei Fei Li)

- AI researcher Fei-Fei Li began working on the idea for ImageNet in 2006.
- In 2007, Li met with Princeton professor Christiane Fellbaum, one of the creators of WordNet to discuss the project.
- They used Amazon Mechanical Turk to help with the classification of images.
- More than 14 million images have been hand-annotated.
- Contains more than 20,000 categories.
- Since 2010, the ImageNet project runs an annual software contest, the ImageNet Large Scale Visual Recognition Challenge (ILSVRC).
- On 30 September 2012, a convolutional neural network (CNN) called AlexNet achieved a top-5 error of 15.3% in the ImageNet 2012 Challenge. DeepLearnig era exploded!!!

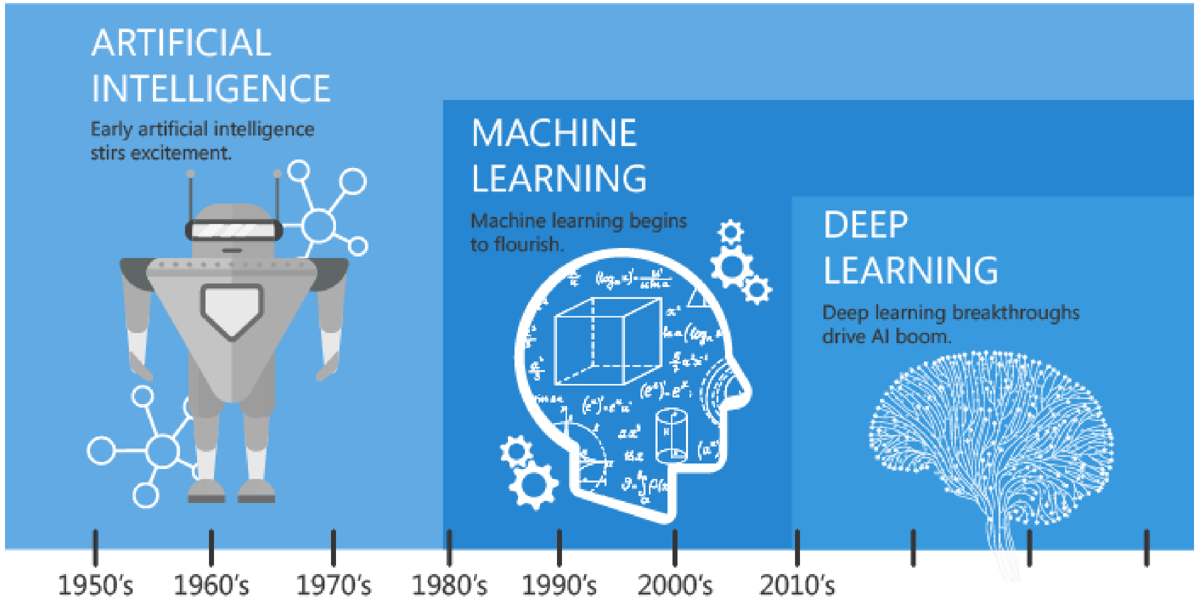
ImageNet results

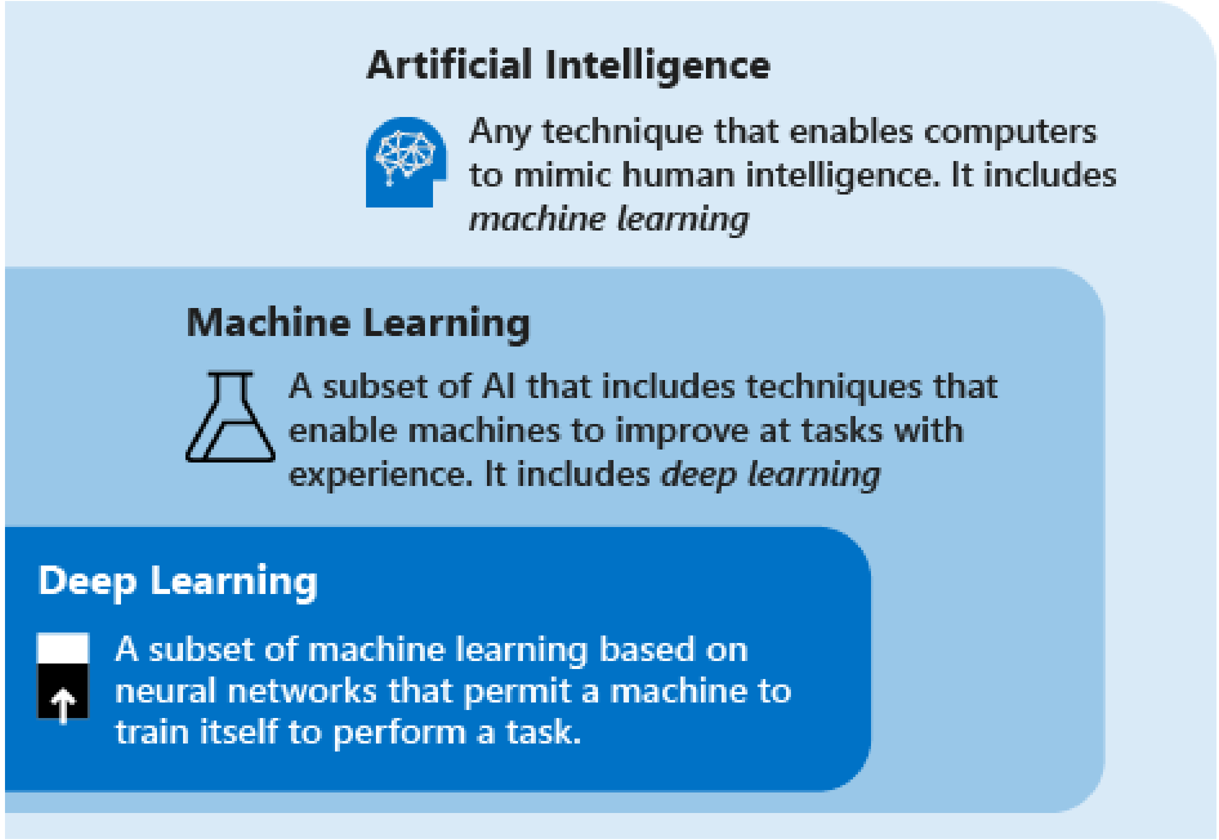


Rank	Method	Top 1 Accuracy	Top 5 Accuracy	Number of params	Extra Training Data	Paper Title	Year	Paper	Code
1	FixResNeXt-101 32x48d	86.4%	98.0%	829M	✓	Fixing the train-test resolution discrepancy	2019		
2	ResNeXt-101 32x48d	85.4%	97.6%	829M	✓	Exploring the Limits of Weakly Supervised Pretraining	2018		
3	ResNeXt-101 32x32d	85.1%	97.5%	466M	✓	Exploring the Limits of Weakly Supervised Pretraining	2018		
4	EfficientNet-B7	84.4%	97.1%	66M	×	EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks	2019		
5	GPIPE	84.3%	97%	557M	×	GPipe: Efficient Training of Giant Neural Networks using Pipeline Parallelism	2018		

https://paperswithcode.com/sota/image-classification-on-imagenet

From AI to ML and DL

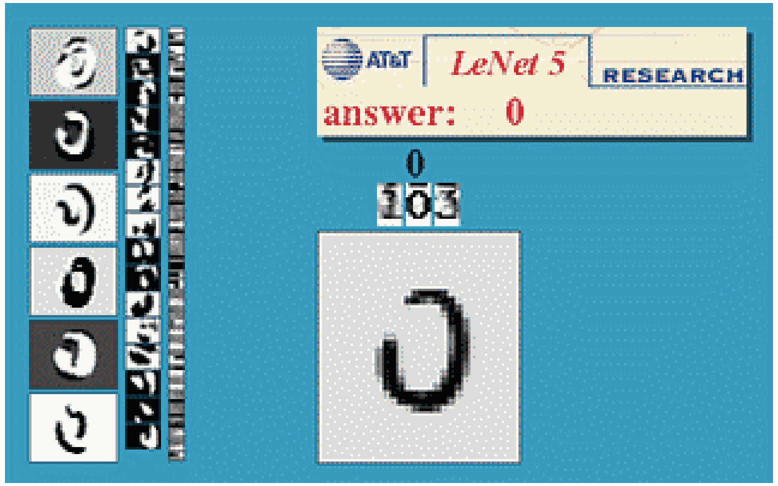




Examples of State-of-the-art

Optical character recognition (OCR)

- Technology to convert scanned docs to text
  - If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs
http://www.research.att.com/~yann/

License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

CAPTCHA

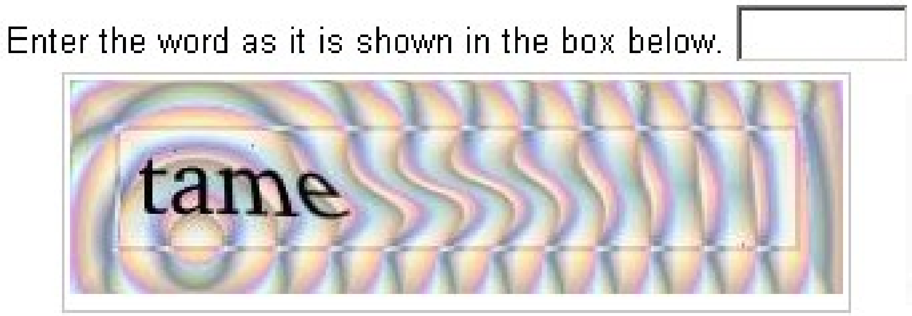
- CAPTCHA stands for "Completely Automated Public Turing test to Tell Computers and Humans Apart".

Breaking visual CAPTCHA results

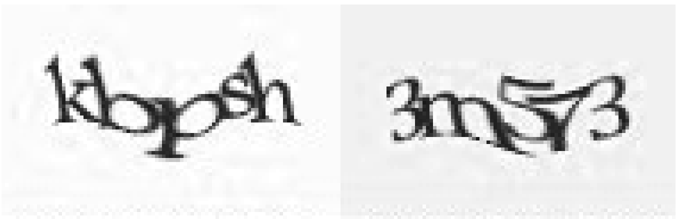
On EZ-Gimpy: a success rate of 176/191=92%!

Other examples

http://www.cs.sfu.ca/~mori/research/gimpy/



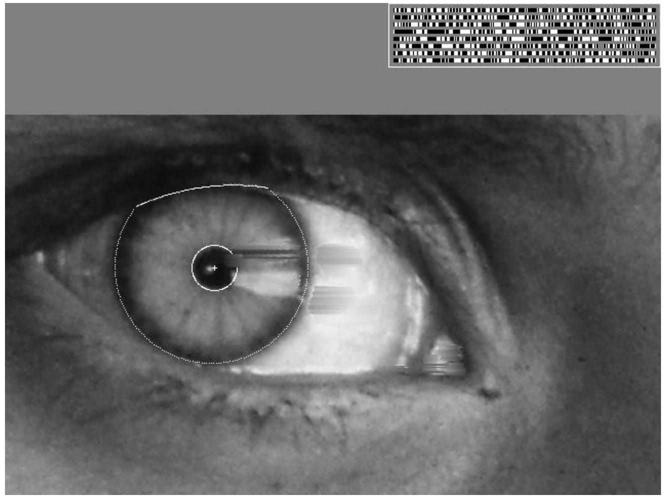
Picture of a CAPTCHA in use at Yahoo.

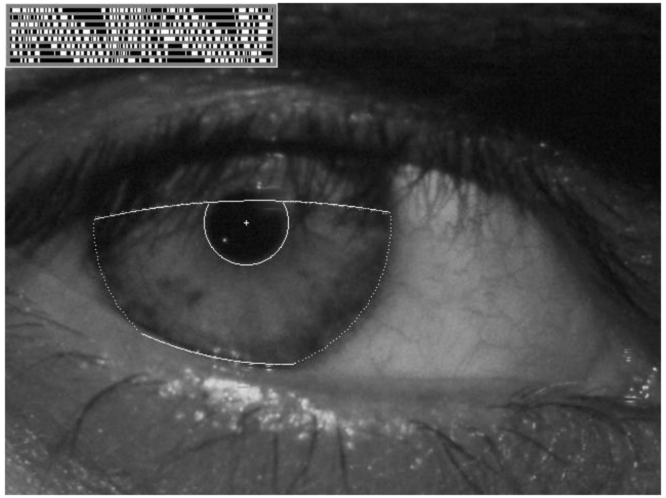


YAHOO's current CAPTCHA format

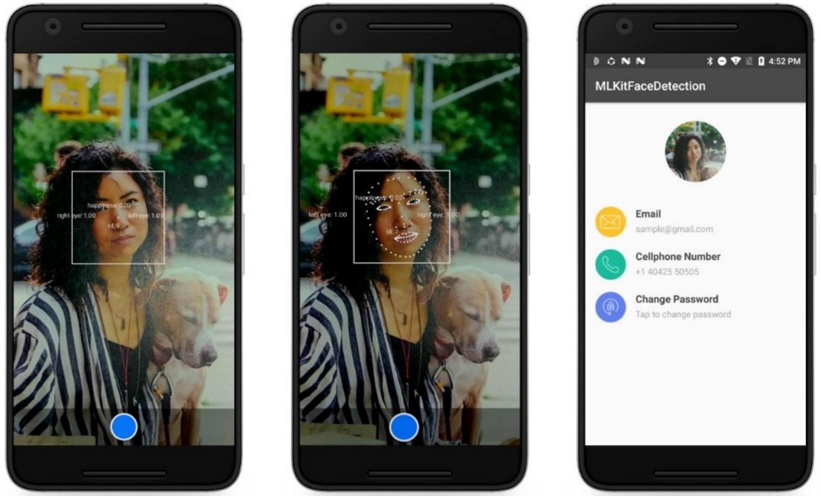
Vision-based biometrics

“*How the Afghan Girl pictured by Steve McCurry in 1984 was Identified by Her Iris Patterns*” confirmed by Daughman by iris recognition (read the story)





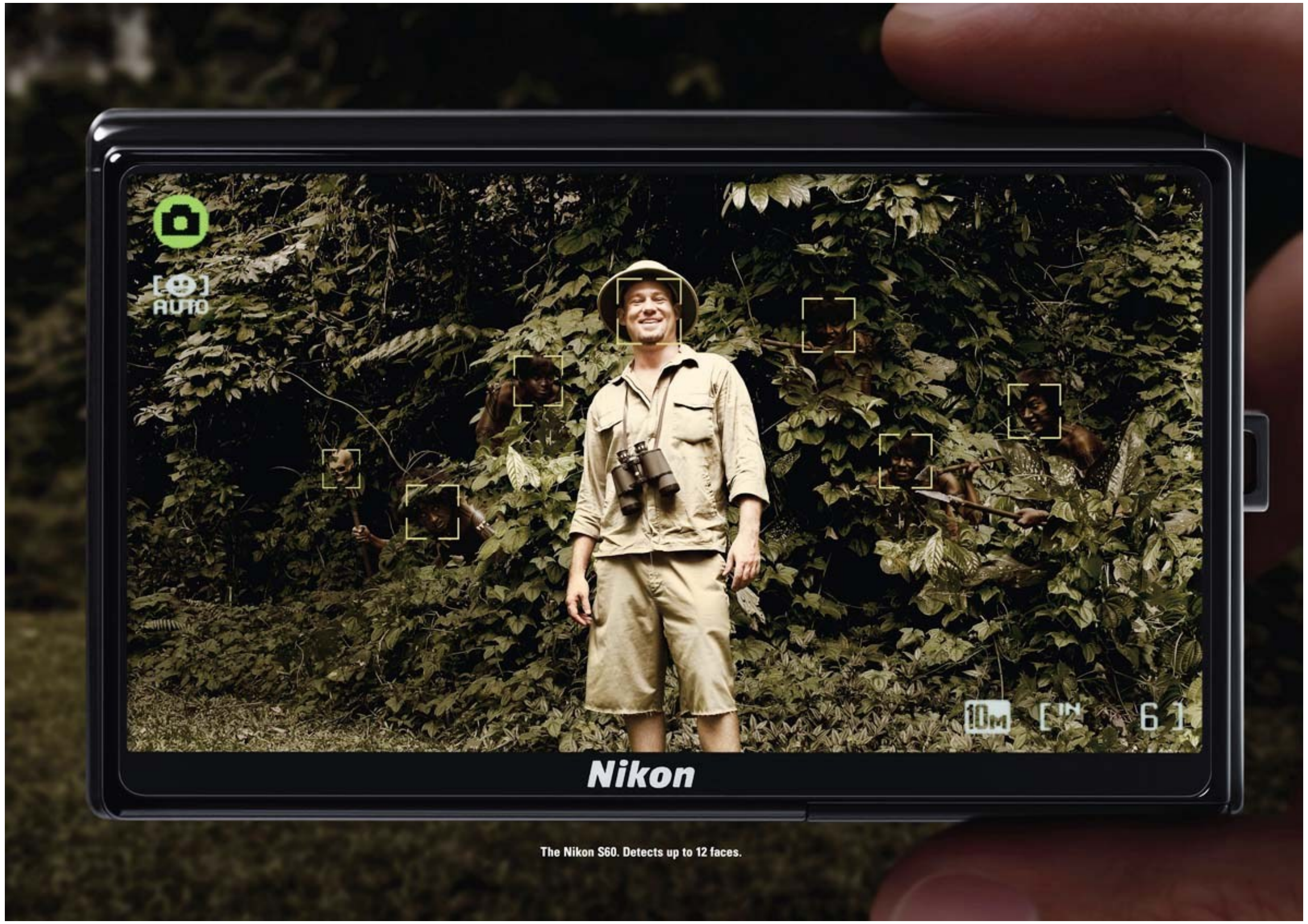
Face detection on digital cameras



Almost all new digital cameras now detect face

- Mobile can also reconstruct the 3D shape of the face!

Still talking about face detection



How many faces can you see in the pictures?

Emotion recognition

[Speech Emotion Recognition with Convolutional Neural Network]

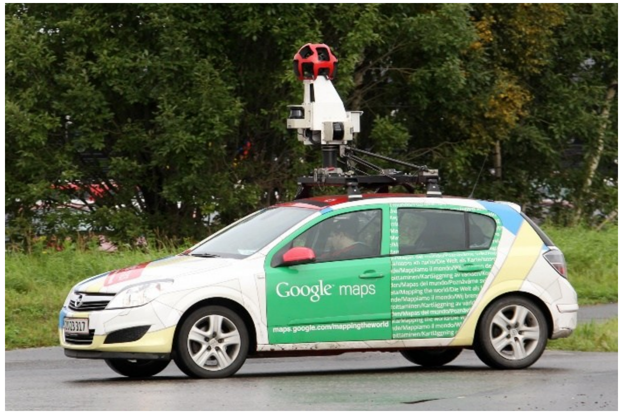
Disgust
Surprise
Sad
Fear
Happy
Contempt

[EmoPy - Python Emotion Recognition Toolkit]

(a) angry (b) disappointed (c) laughing (d) surprised

[Emotion Detection From Facial Expressions (DATA 622)]

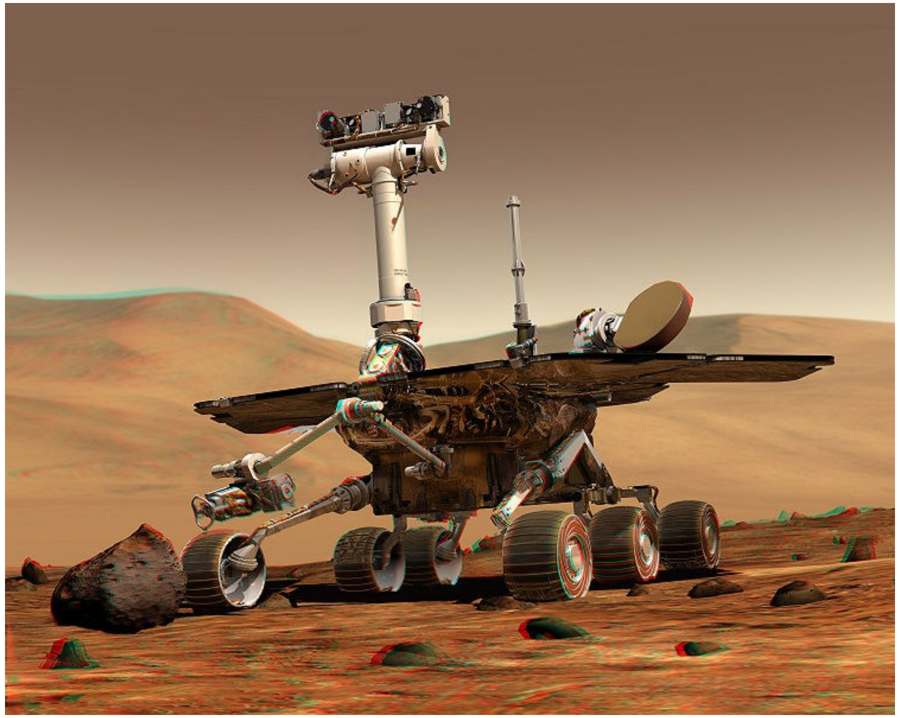
Google cars



Street view

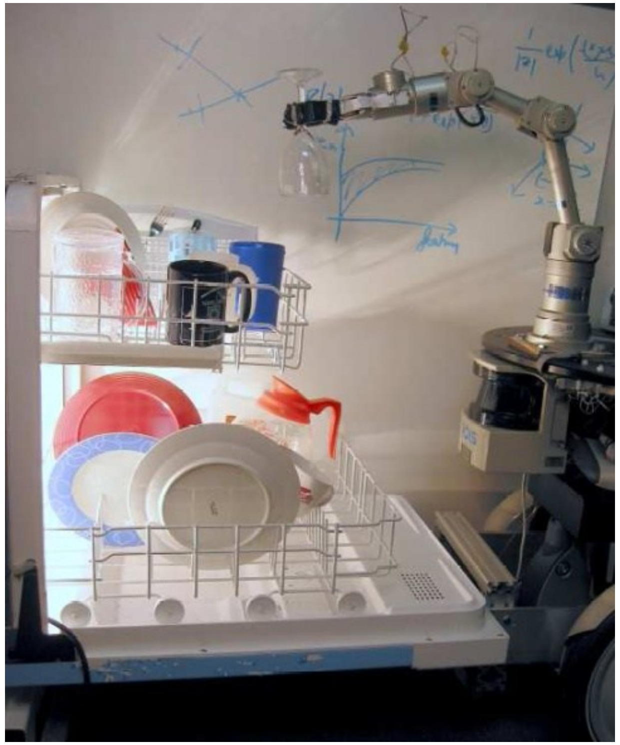
http://www.nytimes.com/2010/10/10/science/10google.html?ref=artificialintelligence

Mobile robots



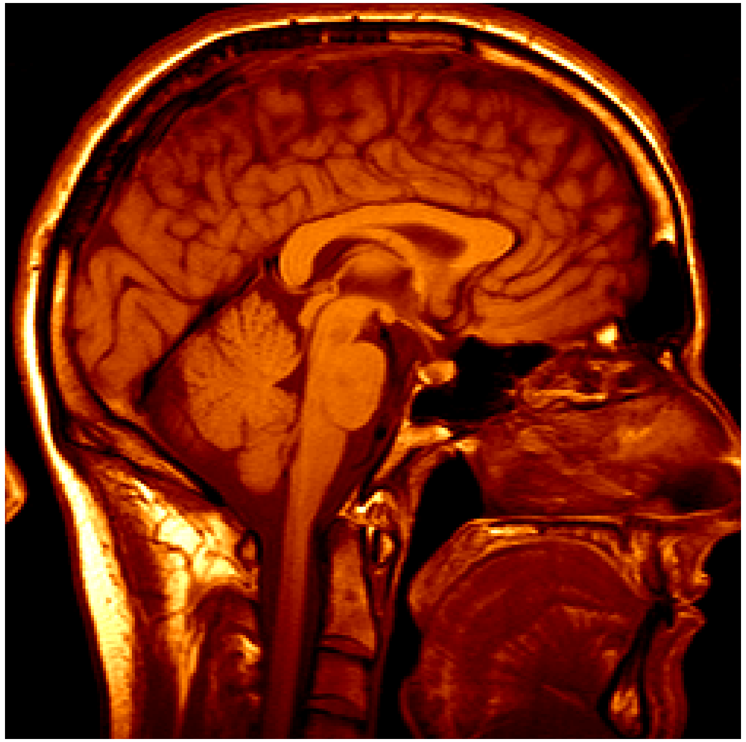
NASA's Mars Spirit Rover
http://en.wikipedia.org/wiki/Spirit_rover

http://www.robocup.org/



Saxena et al. 2008
STAIR at Stanford

Medical imaging



3D imaging
MRI, CT

Image guided surgery
Grimson et al., MIT

C.A.S.A. Course

CONTEXT AWARE SECURITY ANALYTICS IN COMPUTER VISION

proff. Michele Nappi, Fabio Narducci and Carmen Bisogni

aa.2023/2024